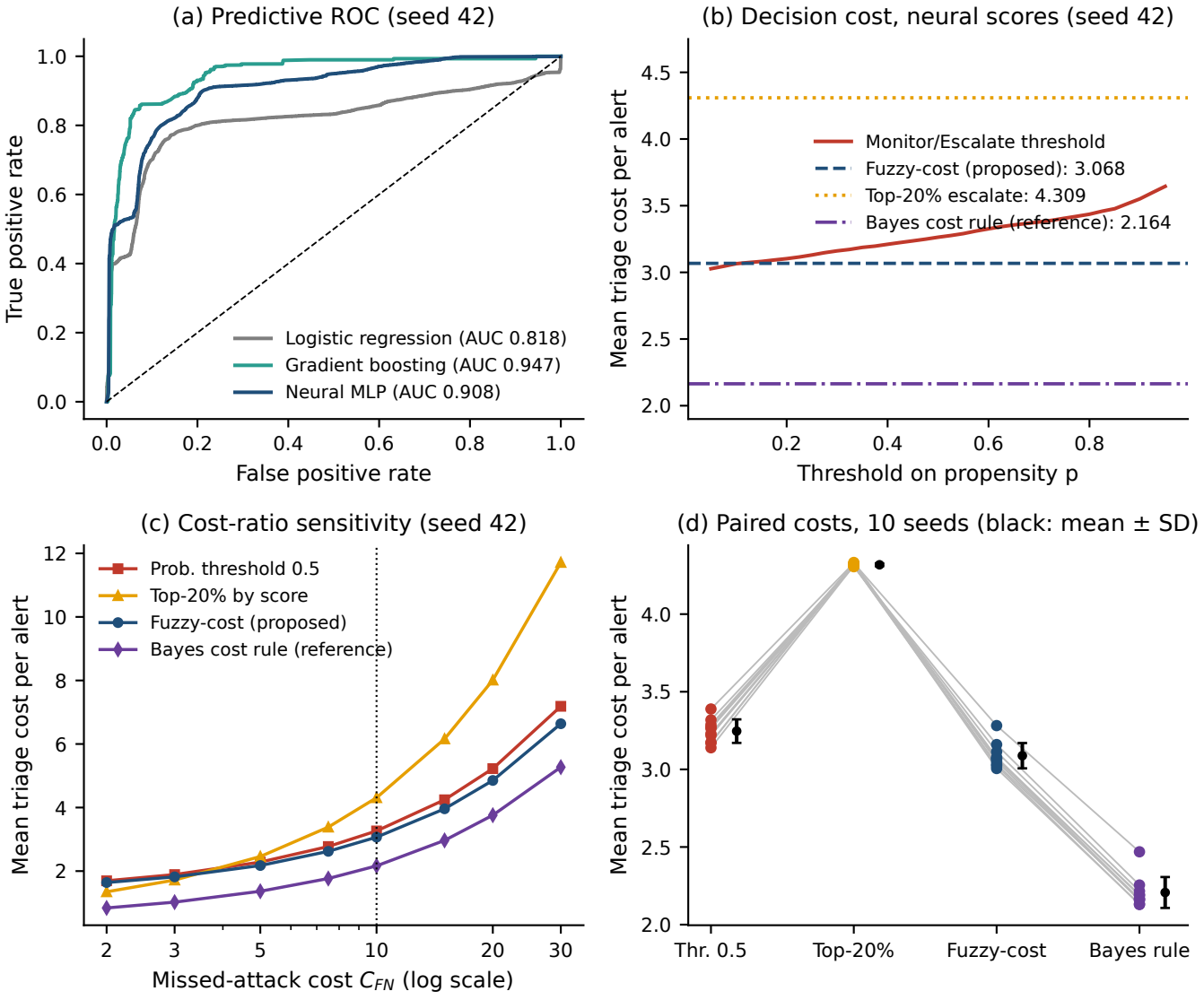
(a) Predictive ROC (seed 42)
True positive rate
False positive rate
Logistic regression (AUC 0.818)
Gradient boosting (AUC 0.947)
Neural MLP (AUC 0.908)
(b) Decision cost, neural scores (seed 42)
Mean triage cost per alert
Threshold on propensity p
Monitor/Escalate threshold
Fuzzy-cost (proposed): 3.068
Top-20% escalate: 4.309
Bayes cost rule (reference): 2.164
(c) Cost-ratio sensitivity (seed 42)
Mean triage cost per alert
Missed-attack cost C_{FN} (log scale)
Prob. threshold 0.5
Top-20% by score
Fuzzy-cost (proposed)
Bayes cost rule (reference)
(d) Paired costs, 10 seeds (black: mean ± SD)
Mean triage cost per alert
Thr. 0.5
Top-20%
Fuzzy-cost
Bayes rule